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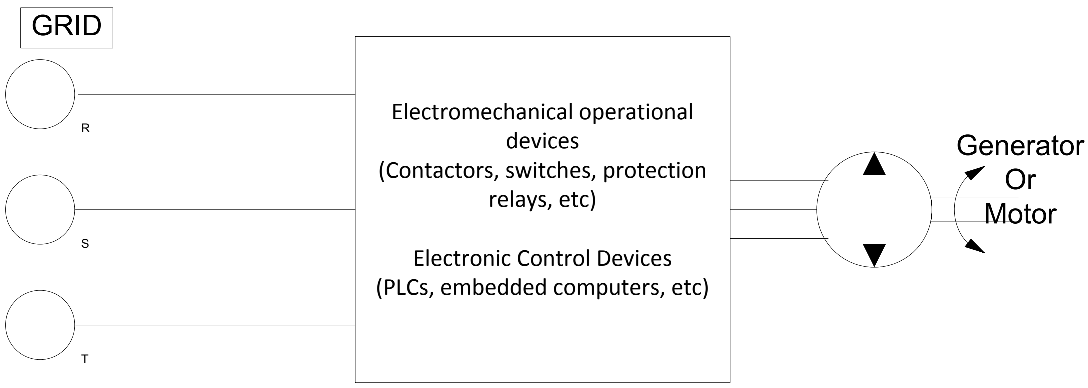
Applications of the GPTR

Contents

The GPTR in an Automatic Test Equipment or Simulation environment

The GPTR is a laboratory programable 3 phase grid generator where you can control the frequency (common), the amplitude of each phase and the phase angle of each phase (2nd and 3rd respect to the first). At the same time it provides 3 current generators, synchronous with the "voltage" generators for each phase, in the range of 0 to 5 amps, where you can control the amplitude of each current generator and the phase angle respect to the first voltage generator. The purpose of these 3 current generators can be better understood considering the applications of the GPTR.

Originally the GPTR was developed for the creation of a real 3 phase grid whose parameters could easily be changed through programming. Consider a simplified connection of a generator or motor or converter to the grid, as in next figure.



That diagram, typically, can be splitted as in next figure.

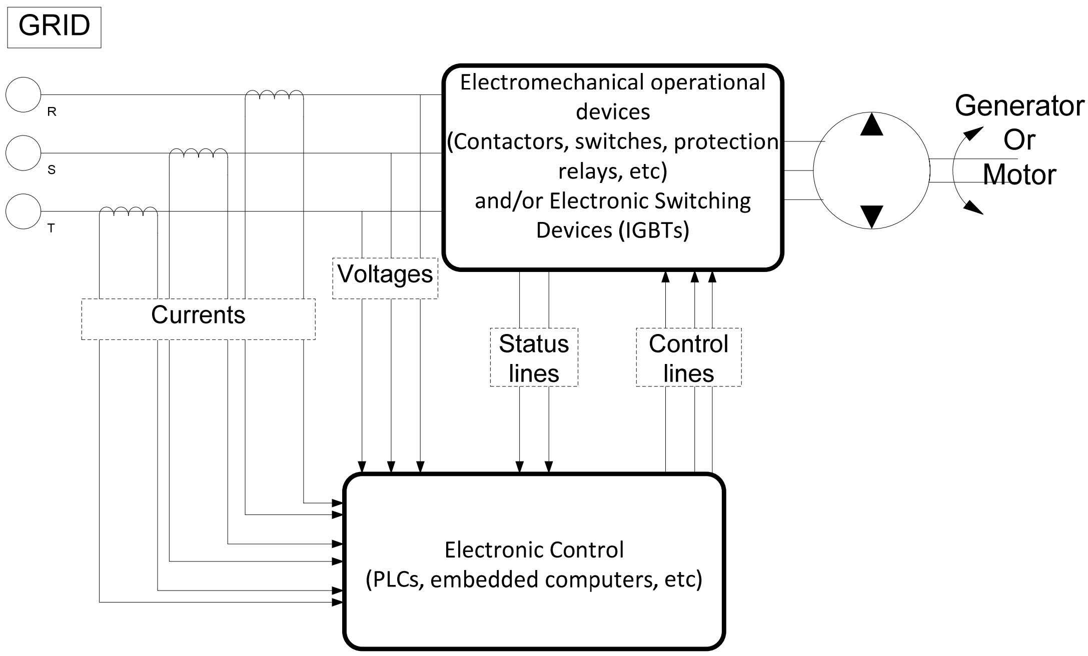


Fig2 – Real environment

Where an "Electronic Control" part receives the status of the elements in the power wires (contactors, IGBTs, etc), and generate signals to control them, while monitoring the overall installation through measuring the voltages in each phase of the grid as well as the current flowing through the power wires.

That schematic applies to any modern installation governing a motor or a generator or a power converter. Having this in mind, next figure shows where the GPTR fits in.

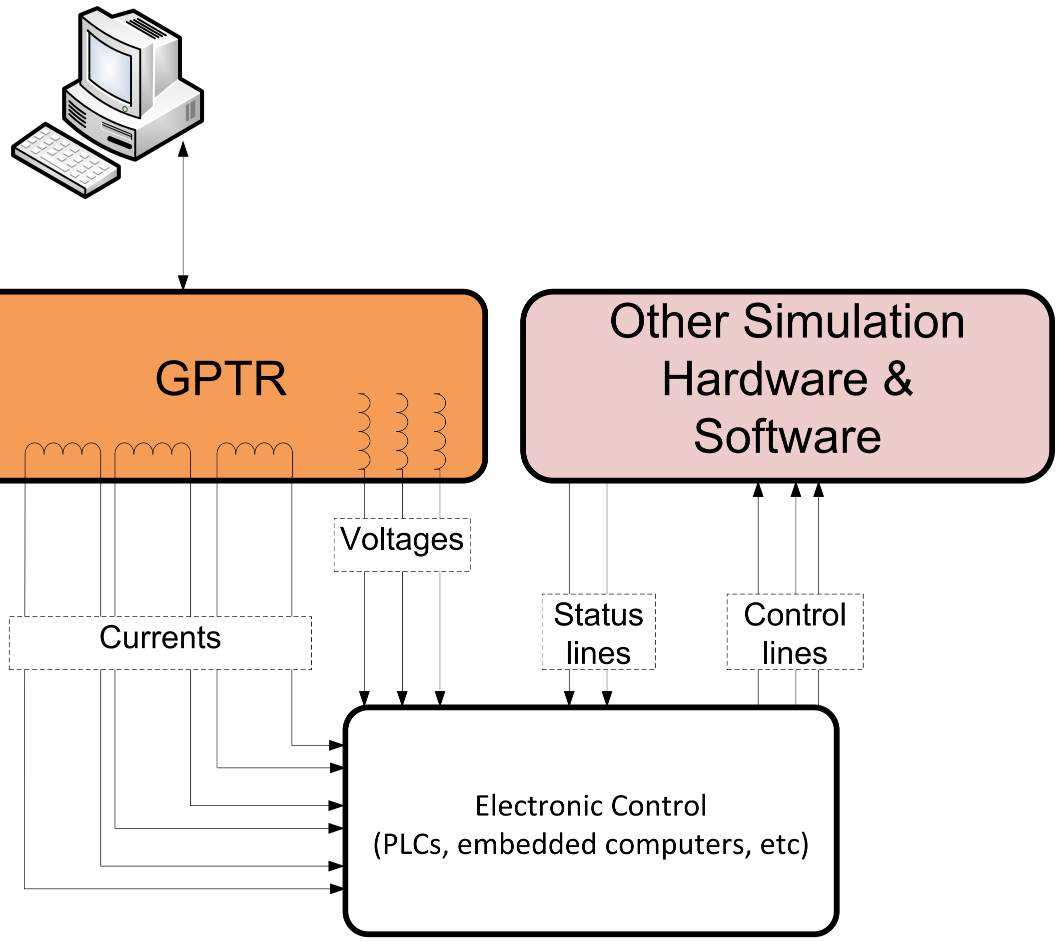


Fig 3 – Test, debugging and validation environment

Typically the "Electronic Control", whether implemented using a PLC, an embedded computer, wired logic, etc, performs sophisticated decisions based of algorithms embedded on it. Actual tests of these algorithms can be quite cumbersome at times, as well as electrical testing of the signals going out of this Control. On top of that, these installations are typically high power and high voltage, so the actual tests in a "real" situation is further complicated due to human security issues, flexibility in the setting of specific test situations, etc.

The signals entering the "Electronic Control" corresponding to the grid voltages have low consumption (just for measuring), sometimes using step-down transformers (in medium voltage installations). The currents flowing through each phase are normally step-down to 1 A or 5 A range using current transformers.

So a solution for the testing of algorithms and its parameters, as well as for functional test of the Electronic Control in these installations can be implemented with a test jig similar to the described in the former figure, where the actual grid is replaced by the GPTR. This solution has a number of advantages:

1. It is very saved environment, as all the lines with high voltage are protected by tripping circuits for high currents.
2. It is very easy to produce conditions and sequences of conditions difficult to generate otherwise.

In the "real" case, the currents in the power lines are supplied by the grid and correspond to the response of the connected power device (generator, motor, etc): i.e., for a given voltage the current in that line will depends on the characteristics of the generator, motor, etc . Instead, with the GPTR the currents actually supplied to the Electronic Control are controlled independently of the voltages and are separated of them, what means that the GPTR can "present" to the Electronic Control an arbitrary load or generation (**four quadrants**).

How the GPTR changes values for voltages and currents?

There are two key concepts to remember:

a) All changes are produced when the corresponding line cross 0 V. This is to minimize the transient effects of switching in the transformer associated to each generator in the GPTR.

b) There is a slope-of-change parameter specific for each generator, both rising and falling.

Similarly when the set value is lower than the actual, the same algorithm is used, but using the values of the slope-of-down-change parameter.

Why the GPTR includes a Grid Analyzer?

To be able to "calibrate" the values of the generators at any moment, after setting these values. See next figure that shows a detailed block diagram.

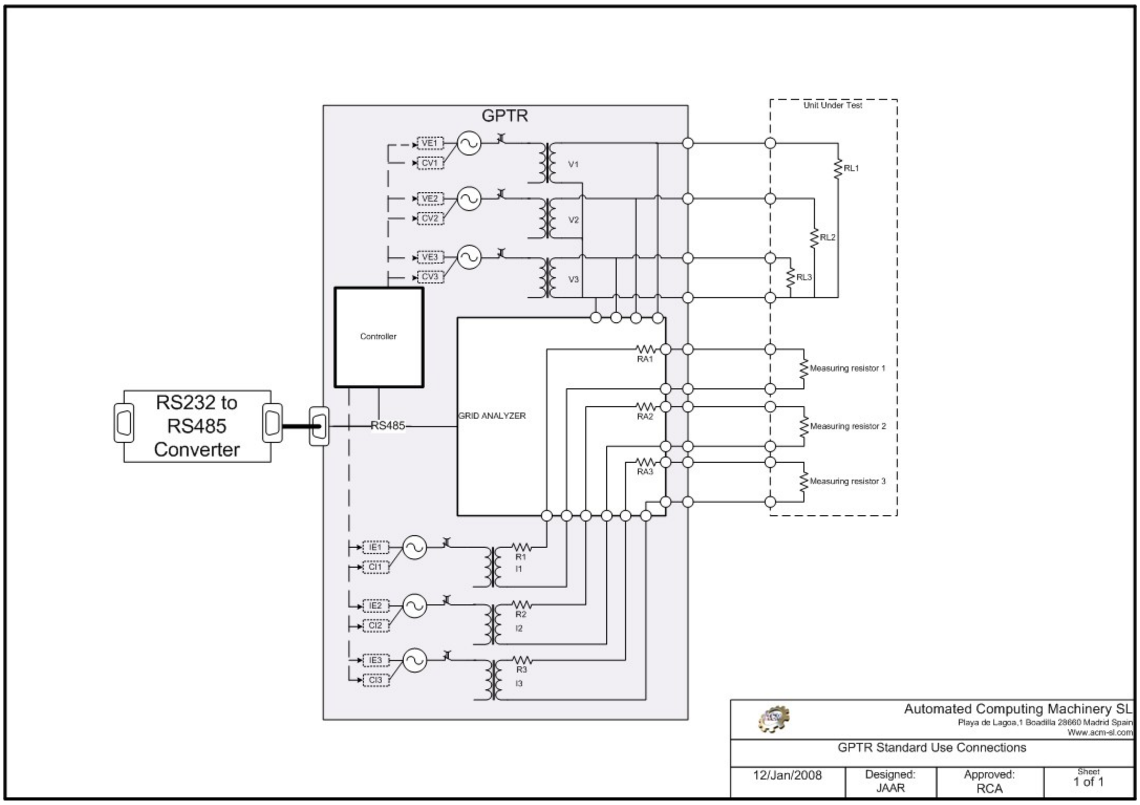
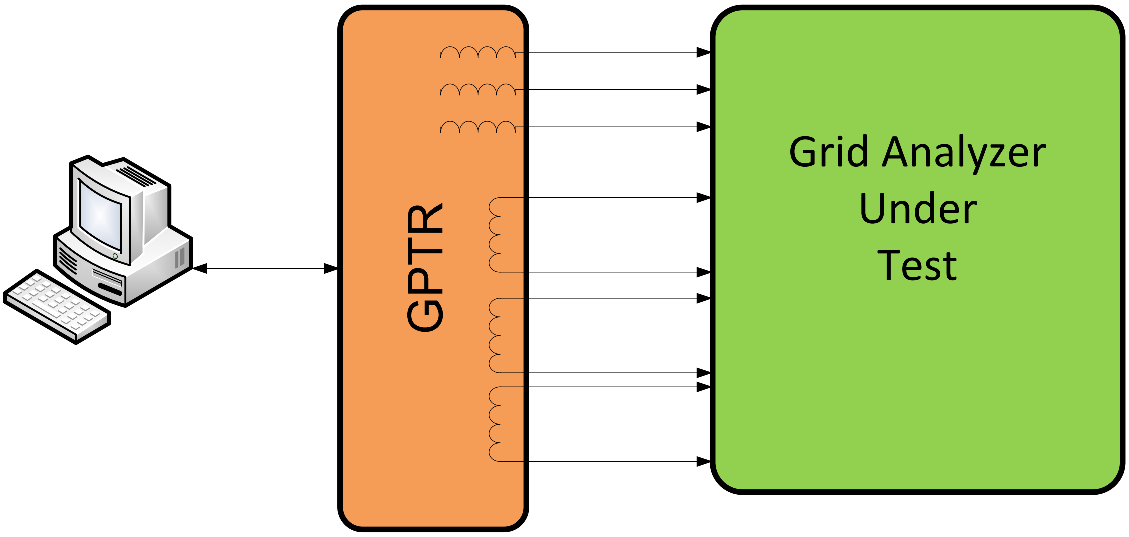


Fig 4 – GPTR block diagram

Calibration of the GPTR generators is done with the command "Calibration" at any moment. In practice you should execute this command only when there has been a change in the load of any generator, either voltage or current.

Use of GPTR for checking Grid Analyzers

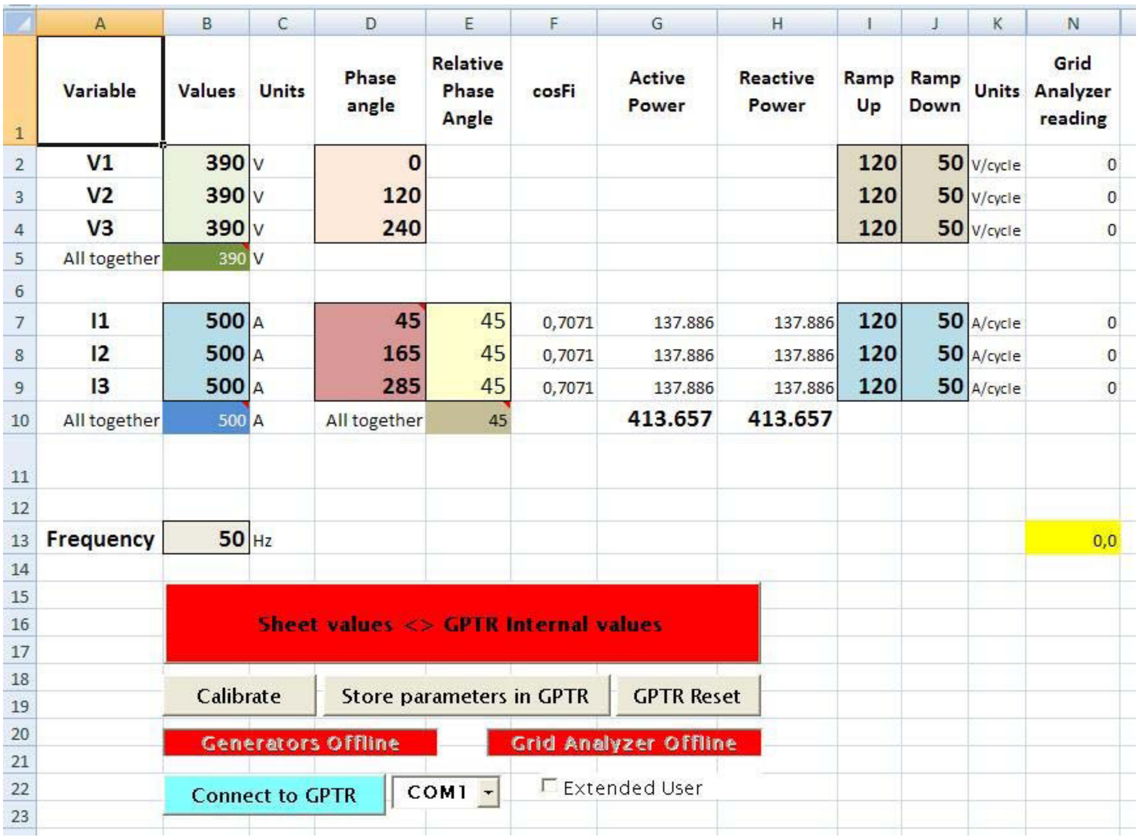
One application for the GPTR is the calibration or checking of calibration of an external Grid Analyzer. (See next figure).



How to control the GPTR?

This is done by programming (there is no manual buttons in the GPTR), through ASCII commands that are sent to the GPTR via a RS485 line.

To ease that programming, the GPTR is supplied with a VisualBasic compatible DLL, hence usable with any VBA programming environment, such as Excel. Also included is an Excel Workbook, with the source code, that can constitutes a very good starting point for the development of a ATE (Automatic Test Equipment) that use the GPTR and, potentially, any other controlling software for the control of additional hardware that may be needed. You can download this Excel WorkBook documentation here ("http://www.acm-sl.com/ingenieria/instrumentation/3-phase-grid-subsystems/gptr-3-phase-grid-generator-and-load-simulator/Excel-GPTR_Command-v1.pdf"). Next figure shows the starting UI for this application.



	A	B	C	D	E	F	G	H	I	J	K	N
1	Variable	Values	Units	Phase angle	Relative Phase Angle	cosFi	Active Power	Reactive Power	Ramp Up	Ramp Down	Units	Grid Analyzer reading
2	V1	390	V	0					120	50	V/cycle	0
3	V2	390	V	120					120	50	V/cycle	0
4	V3	390	V	240					120	50	V/cycle	0
5	All together	390	V									
6												
7	I1	500	A	45	45	0,7071	137.886	137.886	120	50	A/cycle	0
8	I2	500	A	165	45	0,7071	137.886	137.886	120	50	A/cycle	0
9	I3	500	A	285	45	0,7071	137.886	137.886	120	50	A/cycle	0
10	All together	500	A	All together	45		413.657	413.657				
11												
12												
13	Frequency	50	Hz									0,0

This Excel application computes and presents additional information about powers, power factors, status of the communications channel and status of the loads connected to the GPTR. I.e. if a current generator is disconnected from its load, the corresponding cell in the worksheet will change color.

Of course, the user can use all the computing and storage capabilities of Excel to develop:

- Specific sequences for each generator (voltage or current)
- Graphical representation of the results of those sequences.
- Storage of test patterns and their results for further study.